

README FILE

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Course – Fundamentals in Artificial Intelligence and Machine Learning

Project Overview

This project is an AI-based Phishing Email Detector that identifies whether an email/message is phishing i.e. fake or safe.

Initially built using keyword-based detection, I upgraded the project to use Machine Learning (ML) with a real dataset for better accuracy and adaptability.

Features

- Detects phishing emails using Machine Learning.
- Uses TF-IDF vectorization for text processing
- Trained on a real phishing dataset taken from Kaggle.
- Provides:
 - Prediction (Phishing / Safe)
 - Model accuracy
- Simple command-line interface

Project Structure

AIML-Project/

```
|— Data/  
|   └─ Phishing_Email.csv  
|— Phishing_email_detector.py  
|— ProjectReport.pdf  
|— README.md
```

Dataset

- Source: Kaggle (Phishing Email Dataset)
- Contains:
 - Email text/content
 - Labels (Phishing = 1, Safe = 0)
- The dataset is placed inside:
Data/Phishing_Email.csv

Installation

1. Clone the repository
git clone https://github.com/your-username/AIML-Project.git
cd AIML-Project
2. Install dependencies
pip install pandas scikit-learn

How to Run

Run the Python file:

Phishing_email_detector.py

Then enter an email/message:

Enter email/message: Your account is locked. Click here to verify.

Output:

⚠ Phishing Email Detected!

Accuracy: 0.95

How It Works

1. Data Loading

Reads dataset using pandas

2. Text Processing

Converts text into numerical form using:

TF-IDF (Term Frequency - Inverse Document Frequency)

3. Model Training

Uses Logistic Regression to learn patterns

4. Prediction

- Takes user input
- Converts it to vector form
- Predicts phishing or safe

Acknowledgements

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